

Long LI

Applied Mathematician

*Stochastic Modeling and
Scientific Computing*

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Profile

I am an applied mathematician working at the interface of stochastic modeling and scientific computing, with a primary focus on geophysical fluid dynamics. My research develops mathematically grounded and computationally efficient methods for the modeling and simulation of high-dimensional nonlinear systems under uncertainty.

Expertise and Interests: Stochastic Modeling; Uncertainty Quantification; Data Assimilation; Scientific Computing; Numerical Methods; Geophysical Fluid Dynamics; Turbulence

Experience

- 10/2025 – **CNRS Postdoctoral Researcher**, *Institut de Physique de Nice, France*
 - Present
 - Investigating intermittency in scalar turbulence driven by random flows
 - Developing high-fidelity simulations to validate theoretical predictions
 - Extending the framework to multifractal flows via Gaussian multiplicative chaos
- 04/2021 – **Inria Starting Research Positions (Research Fellow)**, *Centre Inria de l'Université de Rennes, France*
 - 07/2025
 - Contributed to the European Research Council (ERC) project **STUOD**
 - Developed stochastic and data-driven models for unresolved subgrid-scale processes
 - Built GPU-accelerated and auto-differentiable solvers with efficient stochastic integrators
 - Studied air-sea interactions and boundary-layer turbulence through stochastic approaches
- 10/2017 – **Doctoral Researcher**, *Institut de recherche mathématique de Rennes, France*
 - 03/2021
 - **PhD Thesis:** *Stochastic modeling and numerical simulation of ocean dynamics*, supervised by Etienne Mémin and Werner Bauer
 - Developed a hierarchy of geophysical models governed by stochastic PDEs
 - Investigated uncertainty quantification and ensemble forecasting in ocean dynamics
- 01/2017 – **Research Intern**, *Laboratoire Jean Kuntzmann, Grenoble, France*
 - 06/2017
 - Master's Thesis: *Global sensitivity analysis for a parametrized diffusion process*, supervised by Clémentine Prieur and Pierre Étoré
 - Developed a novel uncertainty quantification method for stochastic differential equations by combining the Feynman-Kac formula with polynomial chaos expansion.
- 03/2016 – **Research Intern**, *Laboratoire Jean Kuntzmann, Grenoble, France*
 - 06/2016
 - Project: Numerical methods for stochastic differential equations

Education

- 2017 – 2020 **PhD in Applied Mathematics**, *Université de Rennes I, France*
- 2015 – 2017 **Master's Degree in Applied Mathematics**, *Université Grenoble Alpes, France*
- 2012 – 2015 **Bachelor's Degree in Mathematics**, *Université Jean Monnet Saint-Étienne, France*

Supervision and Teaching

- 10/2021 – **Co-supervisor of PhD thesis (50%)**, *Francesco Tucciarone*, Computer Science,
04/2024 University of Rennes 1, France
○ Thesis: *Stochastic parametrization of ocean models through high-resolution observations*
○ Outcome: one journal article and three book chapters
- 04/2024 – **Co-supervisor of Master's internship (50%)**, *Antoine Guines*, Applied Mathe-
06/2024 matics, INSA Rouen, France
○ Subject: *Numerical study of a stochastic ocean model in the Mediterranean configuration*
○ Performed a detailed comparison between deterministic and stochastic simulations
- 03/2015 – **Undergraduate Tutor (30 hours)**, *Faculty of Sciences and Techniques*, Université
05/2015 Jean Monnet, France
○ Led exercise sessions in Analysis and Algebra (first-year level)

Peer Reviewing

- Journals Journal of Advances in Modeling Earth Systems (**JAMES**), Geoscientific Model
Development (**GMD**), Frontiers in Marine Science (**Physical Oceanography**)
- Book Chapters in Mathematics of Planet Earth (**MPE**)

Languages

- Chinese Native
English Fluent (C1)
French Fluent (C1)

Computer Skills

- Programming Python, Fortran, C/C++, PyTorch, Julia, Matlab, R
HPC OpenMP, MPI, CUDA
Tools Vim, Jupyter Notebook, Markdown, LaTeX, ParaView, NetCDF, XIOS
Systems Linux, macOS, Git, SVN, Docker, OAR, SLURM, shell/batch scripting

Selected Software Development

- QGSW** A PyTorch-based solver unifying quasi-geostrophic and shallow-water equations. It is GPU-accelerated, auto-differentiable, and designed for high-resolution and large-ensemble simulations with realistic configurations.
- pyns2d** A PyTorch-based solver for two-dimensional geophysical fluid dynamics, including barotropic quasi-geostrophic and surface quasi-geostrophic models. It provides a flexible platform for turbulence and transport studies.
- QGCM-LU** A stochastic extension of the q-gcm model for representing unresolved subgrid-scale processes in coupled ocean–atmosphere simulations. It also includes tools for coarse-graining and statistical analysis.
- NEMO-LU** A stochastic extension of the NEMO ocean model for improving the representation of unresolved processes in realistic large-scale simulations. It can be used for regional and global ocean forecasting studies.
- SGESM** A stochastic generalized Ekman–Stokes solver for one-dimensional upper-ocean boundary-layer simulations, designed for ensemble simulations and statistical analysis.
- SCESM** An extension of SGESM to coupled ocean–wave–atmosphere dynamics. It is designed to study uncertainty propagation and turbulent transport in coupled boundary-layer processes.

Publications

Journal Articles

- [doi] [pdf] [code] Li, L., Mémin, E., Chapron, B. (2025). A generalized stochastic formulation of the Ekman-Stokes model with statistical analyses. *Journal of Physical Oceanography*, 55, 1389–1407.
- [doi] [pdf] [code] Tucciarone, F., Li, L., Mémin, E., Chandramouli, P. (2025). Derivation and numerical assessment of a stochastic large-scale hydrostatic primitive model. *Journal of Advances in Modeling Earth Systems*, 17, e2024MS004783.
- [doi] [pdf] [code] Clement, S., Blayo, E., Debreu, L., Brankart, J.-M., Brasseur, P., Li, L., Mémin, E. (2025). Link between stochastic grid perturbation and location uncertainty framework. *Journal of Advances in Modeling Earth Systems*, 17, e2024MS004528.
- [doi] [pdf] [code] Thiry, L., Li, L., Mémin, E., Rouillet, G. (2024). A unified formulation of quasi-geostrophic and shallow water equations via projection. *Journal of Advances in Modeling Earth Systems*, 16, e2024MS004510.
- [doi] [pdf] [code] Thiry, L., Li, L., Rouillet, G., Mémin, E. (2024). MQGeometry-1.0: a multi-layer quasi-geostrophic solver on non-rectangular geometries. *Geoscientific Model Development*, 17, 1749–1764.
- [doi] [pdf] [code] Li, L., Deremble, B., Lahaye, N., Mémin, E. (2023). Stochastic data-driven parameterization of unresolved eddy effects in a baroclinic quasi-geostrophic model. *Journal of Advances in Modeling Earth Systems*, 15, e2022MS003297.
- [doi] [pdf] [code] Brecht, R., Li, L., Bauer, W., Mémin, E. (2021). Rotating shallow water flow under location uncertainty with a structure-preserving discretization. *Journal of Advances in Modeling Earth Systems*, 13 (12), e2021MS002492.
- [doi] [pdf] Bauer, W., Chandramouli, P., Chapron, B., Li, L., Mémin, E. (2020). Deciphering the role of small-scale inhomogeneity on geophysical flow structuration: a stochastic approach. *Journal of Physical Oceanography* 50, 983-1003.
- [doi] [pdf] Bauer, W., Chandramouli, P., Li, L., Mémin, E. (2020). Stochastic representation of mesoscale eddy effects in coarse-resolution barotropic models. *Ocean Modelling* 151, 101646.
- [doi] [pdf] [code] Resseguier, V., Li, L., Jouan, G., Derian, P., Mémin, E., Chapron, B. (2020). New trends in ensemble forecast strategy: uncertainty quantification for coarse-grid computational fluid dynamics. *Archives of Computational Methods in Engineering*, 1886-1784.
- [doi] [pdf] Étoré, P., Prieur, C., Pham, D., Li, L. (2020). Global sensitivity analysis for models described by stochastic differential equations. *Methodology and Computing in Applied Probability* 22, 803-831.

Book Chapters

- [doi] [pdf] [code] Li, L., Mémin, E., Chapron, B. (2026). A stochastic Ekman-Stokes model for coupled ocean-atmosphere-wave dynamics. *Stochastic Transport in Upper Ocean Dynamics IV, Mathematics of Planet Earth*, vol 15. Springer, 105-131.
- [doi] [pdf] Tucciarone, F., Li, L., Mémin, E., Thiry, L. (2025). Transport noise defined from wavelet transform for model-based stochastic ocean models. *Stochastic Transport in Upper Ocean Dynamics III, Mathematics of Planet Earth*, vol 13. Springer, 287-303.
- [doi] [pdf] [code] Mémin, E., Li, L., Lahaye, N., Tissot, G., and Chapron, B. (2024). Linear wave solutions of a stochastic shallow water model. *Stochastic Transport in Upper Ocean Dynamics II, Mathematics of Planet Earth*, vol 11. Springer, 223–245.
- [doi] [pdf] Tucciarone, F., Mémin, E., Li, L. (2024). Data driven stochastic primitive equations with dynamic modes decomposition. *Stochastic Transport in Upper Ocean Dynamics II, Mathematics of Planet Earth*, vol 11. Springer, 321–336.
- [doi] [pdf] Jamet, Q., Mémin, E., Dumas, F., Li, L., Garreau, P. (2024). Toward a stochastic parameterization for oceanic deep convection. *Stochastic Transport in Upper Ocean Dynamics II, Mathematics of Planet Earth*, vol 11. Springer, 143–157.

- [[doi](#)] [[pdf](#)] [[code](#)] Li, L., Mémin, E., Tissot, G. (2023). Stochastic parameterization with dynamic mode decomposition. *Stochastic Transport in Upper Ocean Dynamics, Mathematics of Planet Earth*, vol 10. Springer, 179–193.
- [[doi](#)] [[pdf](#)] [[code](#)] Thiry, L., Li, L., Mémin, E. (2023). Modified (hyper-)viscosity for coarse-resolution ocean models. In *Stochastic Transport in Upper Ocean Dynamics*, 273-285. Springer, Cham.
- [[doi](#)] [[pdf](#)] Tucciarone, F., Mémin, E., Li, L. (2023). Primitive equations under location uncertainty: analytical description and model development. *Stochastic Transport in Upper Ocean Dynamics, Mathematics of Planet Earth*, vol 10. Springer, 287–300.
- [[doi](#)] [[pdf](#)] Fiorini, C., Boulevard, P.-M., Li, L., Mémin, E. (2023). A Two-step numerical scheme in time for surface quasi-geostrophic equations under location uncertainty. *Stochastic Transport in Upper Ocean Dynamics, Mathematics of Planet Earth*, vol 10. Springer, 57–67.

Presentations

Invited Talks

- [[slides](#)] Li, L. (Mar. 2025). Stochastic modeling in geophysical fluid dynamics. ANGE Project-Team Seminar, Sorbonne Université, Paris, France.
- [[slides](#)] Li, L. (Oct. 2024). Stochastic modeling in geophysical fluid dynamics. CALISTO Project-Team Seminar, Centre Inria d'Université Côte d'Azur, Valbonne, France.
- [[slides](#)] Li, L. (Sep. 2024). Stochastic modeling in geophysical fluid dynamics. Seminar at Research Center for Mathematics and Interdisciplinary Sciences, Shandong University, Qingdao, China.
- [[slides](#)] Li, L. (Nov. 2022). Stochastic transport in an idealized ocean-atmosphere coupled model. AIRSEA Project-Team Seminar, Centre Inria de l'Université Grenoble Alpes, France.

Selected Talks

- [[slides](#)] Li, L. (Sep. 2025). Stochastic approaches to air–sea boundary layer dynamics. 6th STUOD Annual Workshop, ICMS The Bayes Centre, Edinburgh, UK.
- [[slides](#)] Li, L. (Oct. 2024). A stochastic formulation of the Ekman-Stokes model. Atelier sur la représentation des fines échelles océaniques dans les simulations numériques, Ifremer, Plouzané, France.
- [[slides](#)] Li, L. (Sep. 2024). A generalized stochastic formulation of the Ekman-Stokes model with statistical analyses. 5th STUOD Annual Workshop, Inria Rennes, France.
- [[slides](#)] Li, L. (Sep. 2023). Uncertainty quantification of mesoscale air-sea interaction under transport noise. 4th STUOD Annual Workshop, Ifremer, Plouzané, France.
- [[slides](#)] Li, L. (Sep. 2022). Stochastic transport in an idealized ocean-atmosphere coupled model. 3rd STUOD Annual Workshop, Imperial College London, UK.
- [[slides](#)] Li, L. (May 2022). Stochastic transport in an idealized ocean-atmosphere coupled system. EGU General Assembly, Vienna, Austria.
- [[slides](#)] Li, L. (Apr. 2021). Stochastic modeling of mesoscale eddies. EGU General Assembly.
- [[slides](#)] Li, L. (Sep. 2019). Stochastic modeling of mesoscale eddies in oceanic dynamics. Workshop on Frontiers of Uncertainty Quantification in Fluid Dynamics, Pisa, Italy.
- [[slides](#)] Li, L. (Apr. 2019). Oceanic dynamics under Location Uncertainty - Towards a consistent stochastic modeling. Workshop Conservation Principles, Data and Uncertainty in Atmosphere-Ocean Modeling, Potsdam, Germany.